Week 8
Lecture 10
DATA GATHERING II

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Ontology vs Epistemology in Research

- Ontology: addresses "what is"
- Epistemology: addresses "how do we know“

- Ontology: deals with reality of entities
- Epistemology: deals with methods of acquiring knowledge

- Ontology: concerned with categorizing entities
- Epistemology: concerned with evaluating evidence

Ontology vs Epistemology in Research

- Ontology in UX research: determining if a website is considered a single entity or multiple entities
- Epistemology in UX research: determining if usability testing is the most effective method for evaluating user satisfaction with a website

- Ontology in UX research: determining if chatbot is considered as human or machine
- Epistemology in UX research: determining if A/B testing is an appropriate method for evaluating the effectiveness of a chatbot's conversational design

What is Qualitative Research?

- Qualitative research involves collecting and analyzing non-numerical data (e.g., text, video, or audio) to understand concepts, opinions, or experiences. It can be used to gather in-depth insights into a problem or generate new ideas for research.

https://www.scribbr.com/methodology/qualitative-research/

Nuances of Qualitative Research

- Data interpretation needed
- Observation/measurement changes phenomenon
- Numbers, equations, words are symbolic, not reality
- Validity is a goal, not a product

Issues with Qualitative Research

Reliability in Research (1)

- Validity, reliability and generalisability have been called the ‘holy trinity’ of the sciences (Kvale 1996).

Reliability in Research (2)

- Reliability: consistency of research findings
- Repeatability of results under same conditions
- Reduces random errors, chance influences
- Two types: test-retest, inter-rater
- Test-retest: consistency over time on same individuals
- Inter-rater: consistency of results among multiple judges
- Ensures **accuracy and trustworthiness** of findings.

Validity in Research

- Validity: accuracy of research findings
- Ensures **conclusions are accurate** and meaningful
- Ensures findings can be replicated, **generalizable**
- Two types: internal and external
- Internal: findings accurately reflect phenomenon
- External: findings generalizable to other settings/populations
- Important for reliability and trustworthiness of findings.

Internal Validity for UX Studies

- Study compares two sites (A and B)
- Participants shown site A first, then site B
- Same tasks completed on both sites
- This study design may not produce accurate results
- May not accurately identify the better design

Internal Validity for UX Studies

- Study setup favors design B
- Participants already familiar with testing situation and task domain
- May have expectations and know what's expected of them
- Lacks internal validity
- Solution is to alternate which site goes first and have half of the users try site B first

External Validity for UX Studies

A study has external validity if the participants and the study setup are representative for the real-world situation in which the design is used but:

- Recruiting study participants from the general population may not be valid for seniors
- Testing mobile design on desktop may not generalize to real-world use
- Studies may lack external validity
- Further study may be required to confirm results.

Recommendations for Study Design: Internal Validity

- Randomize task order to avoid bias
- People take longer and make more errors on first tasks
- Tasks at end of session may show effect of fatigue
- Use warmup tasks to mitigate learning phase and make participants comfortable
- Ensure warmup tasks are not included in analysis
- Sometimes, task order may not be feasible to randomize

Recommendations for Study Design: External Validity

- Recruit representative participants matching target audience demographics and goals
- External validity may not be ensured by demographics alone
- Strive to find participants with similar goals as users
- Replicate natural usage environment for best results
- Environment plays role in usability, test in location it will be used.

Reactivity in Research

- Reactivity refers to change in behavior due to observation
- Can occur in quantitative and qualitative research
- Examples include Hawthorne effect, conformity effect, and social desirability bias
- Can affect research validity and generalizability.

Reactivity in Research: Hawthorn Effect

- Hawthorne effect: phenomenon where participants change behavior/performance due to increased attention in study
- Named after experiments at Hawthorne Works in 1920s/1930s
- Can lead to inaccurate results if not controlled for in study design.

Reactivity in Research: Conformity Effect

- Conformity effect refers to the tendency for individuals to conform to the actions or opinions of a group, in order to fit in or be liked by others.
- It can occur in both quantitative and qualitative research.
- Examples include the Asch conformity experiments, where participants conformed to the incorrect answer given by a group, and groupthink, where individuals conform to group decision making without critically evaluating the decision.
- Conformity effect can lead to inaccurate research results if not controlled for in the study design.

Reactivity in Research: Conformity Effect

- https://www.youtube.com/watch?v=TrTk6DsEJ2Q&tab_channel=HappinessAccelerator

Reactivity in Research: Conformity Effect

- Blind study design: Participants and/or researchers do not know the group assignments (control or experimental group)
- Double-blind study design: Both Participants and researchers do not know the group assignments.
- Helps to control for observer expectancy effect by reducing researcher bias.
- Commonly used in clinical trials to control for placebo effect.

Reactivity in Research: Social Desirability Effect

- Social desirability effect: phenomenon where participants give responses that they believe are socially acceptable or desirable, rather than their true beliefs or behavior.
- Can occur in both quantitative and qualitative research.
- Examples include participants over-reporting positive behaviors or under-reporting negative behaviors.
- Can lead to inaccurate research results if not controlled for in the study design.
- Anonymous surveys or indirect questioning methods can help control for this effect.

Reactivity in Research: Social Desirability Effect

- https://www.youtube.com/watch?v=J930Rut_J6E&tab_channel=NationalGeographic

Sampling

- Sampling: selecting a subset of a population to study
- Representativeness: ensuring sample is representative of population
- Sample size: determining appropriate number of individuals to include
- Sampling bias: potential for sample to not accurately represent population

Sampling

- Quant research: sample characteristics, size more important than relative size
- Qual research: not statistical representativeness, but full range of views, themes and possibilities.

Sampling

- The appropriate sample design will depend on the following criteria:
  - Degree of accuracy required
  - Nature of the project local versus national project
  - Need for statistical analysis
  - Resources (i.e. time / money)

Sampling Categories

- Probability sampling: random selection of individuals
- Non-probability sampling: non-random selection of individuals ex. quota sampling
- Convenience sampling: selecting individuals easily accessible
- Stratified sampling: dividing population into subgroups and selecting from each
- Cluster sampling: dividing population into clusters and randomly selecting clusters
- Snowball sampling:

  recruiting participants through referrals

Promoting Validity & Reliability in Research

- Credibility: **Active listening**, reflection
- Confirmability: Direct participatory, **documented evidence** from primary sources
- Meaning-in-context: Understanding data in **holistic contexts**
- Recurrent patterning: Identifying patterns through **repeated experiences**
- Transferability**: Examining general similarities** of findings in similar situations

Leininger 1994

Promoting Validity & Reliability in Research

- Credibility
- Applicability
- Consistency
- Neutrality

Lincoln and Guba 1990

Promoting Validity & Reliability in Research

- Member checks: Confirming data and interpretations with participants
- Also called member tests of validity, host verification
- Can be conducted continuously and both formally and informally
- Seeking diversity
- Allows for range of realities and perspectives
- Examples: Peer reviews, diary keeping, discussions with colleagues etc.

Merriam 2002

Promoting Validity & Reliability in Research

- Audit Trails: Detailed account of methods, procedures, decision points
  - Raw data
  - Data reduction/analysis products
  - Data reconstruction/synthesis products
  - Process notes (journal)
  - Materials on intentions/dispositions

Merriam 2002

Audit Trail: An Example

Operation	Time	Sender / Operator	Recipients	Content
email	09/02/03 00:00:00	Zhang	M. Johnson	Hi Mike, our paper has been accepted …
doc	09/03/04 01:01:01	Johnson		The purpose of this travel is to learn the latest development in …
email	09/03/05 09:09:09	Johnson	S. Bergman	Hi Sarah, In August, I will have a trip to Boston for a conference …
web	09/03/06 09:09:09	Bergman		Air Canada ticket center…

https://link.springer.com/chapter/10.1007/978-3-642-04125-9_47

Ethics in Qualitative Research

- Validity and reliability depend on **investigator's ethics**
- Ethical practice based on **individual researcher's** values and ethics
- Protection of subjects, informed consent, privacy, and deception must be considered
- Credibility of **researcher**, rigorous methods, and understanding of qualitative inquiry essential for credibility of qualitative research
- Credibility includes: training, experience, track record, status, self-presentation, intellectual rigor, professional integrity, and methodological competence
- Trust in study's integrity and researcher's ethical stance necessary.

Ethical Issues Checklist

- Patton (2002) identifies the following ten items to be considered when engaging in qualitative research:

1. Explaining purpose of the inquiry and methods to be used
2. Promises
3. Risk assessment
4. Confidentiality
5. Informed consent
6. Data access and ownership
7. Interviewer mental health
8. Advice (who will be your counselor on ethical matters)
9. Data collection boundaries
10. Ethical versus legal conduct

Ethical Guidelines

- Research Ethics Board **approval** may be required for certain types of research, and a detailed research plan is necessary for approval
- Research participants often expect their **anonymity** to be protected
- Ethics **guidelines** may be **formal or informal**, depending on the industry
- Research ethics concerns focus on **risk and consent**

Risk

- Research should minimize the potential for harm to participants
- In medical research, harm is often clear
- In user research, it can be more difficult to anticipate harm
- Identities of research participants are often undisclosed to protect against potential harm
- Research can unintentionally do harm through disclosures made during the research process
- It is important to consider potential harm when conducting research.

Privacy and Confidentiality

- Privacy and confidentiality are important considerations in research ethics
- Researchers should ensure that research participants cannot be identified from their responses
- This is important for both published and stored data
- Numeric codes or aliases may be used to protect participants' identities
- The more personal or "human" the research, the higher the risk of participants being identified by their responses
- Researchers should consider the potential for unintended harm through disclosures during research

Consent

- Research involving human participants must follow the principle of informed consent
- Participants must be fully aware of potential risks and understand the extent of those risks
- Risks cannot be hidden or minimized, even if they are minor
- In research where participants may reasonably expect harm, the benefits must outweigh the risks
- Participants must fully understand the risks and agree to take them